Economics 8803: Economic Data Analysis

Problem Set 4

Due by: **April 14, 5 PM ET**

Instructions:

- Same as in previous Problem Sets.

1 Empirical Question

This problem set is based on the paper by Anderson (JHE 2008) and, is a modified version of an earlier problem set by Michael Anderson. While not the main focus of that paper, it collected data on primary and secondary seat belt laws. A secondary belt law stipulates that law enforcement can only ticket a driver for not wearing a seat belt if the driver has already been pulled over for simultaneously breaking a different traffic law. A primary belt law stipulates that law enforcement can ticket a driver for not wearing a seat belt, regardless of whether they have broken any other laws. **This problem set will focus on the question of whether primary seat belt laws save lives.**

Unlike the data from the problem set 2, the state-by-year panel here has already been cleaned. The labels on the variables should be self-explanatory. As before, the data in DTA and CSV formats (`pset4.csv` and `pset4.dta`) can be downloaded from Canvas, and the TXT file gives variable names and labels.

1. Data summary and pre-analysis:

   (a) Start by summarizing the data (Hint: package `panelview` in Stata and `R` can be helpful in both parts): Is the panel balanced (a.k.a. complete)? Visualize the timing of primary belt laws. Are there any reversals of primary belt laws? Are there never-treated states?

   (b) Note that the data also includes information on the secondary seat belt laws. Use `panelview` to visualize the timing of primary and secondary belt laws in the data. What do you observe?

   (c) Define the outcome as log traffic fatalities per capita. What would be problematic if we measured the outcome as just the number of traffic fatalities?

(d) State the parallel trends assumption in words (try to be as precise and concise as you can and please avoid simply translating the math expression in words).

(e) Plot raw outcome data in a way that may be helpful for later difference-in-differences (DiD) analysis (`panelview` may come in handy here too). From this graph, can you say anything about the plausibility of the parallel trends and no anticipation assumptions, as well as the effects of primary belt laws?

(f) Now, conduct a test for anticipation effects. One way of doing this is to run a two way fixed effects regression on the untreated data only. The coefficients of interest will be time period indicators leading up to the treatment. Show these coefficients along with the appropriate standard errors in a table or as a figure. What do you observe?

2. Difference-in-Differences (DID) analysis:
You may use canned commands to implement DID or code it manually. For the event study specifications, report the estimates visually along with 95% confidence bands.

   (a) Use the static Two Way Fixed Effects (TWFE) to estimate the treatment effect of primary belt law. Interpret the treatment effect.

   (b) Refer to your summary from `panelview` for part 1(a), and list any three examples of 2×2 comparisons that the TWFE specification you estimated above performs. Briefly state, which comparisons could be problematic?

   (c) Use Callaway and Sant'Anna (2021) group-time average estimator to estimate the event study specification. Use the 'never-treated group' as the control group for the treated units. What is the overall ATT estimate? Are the estimates different if you also use the 'not-yet-treated' group as controls?

   (d) Implement the Borusyak, Jaravel, and Spiess (2022) (BJS) imputation estimator for the event-study specification. How does the event-study from BJS compare with the event-study from Callaway and Sant'Anna estimator?

3. Oftentimes authors conduct several robustness checks and placebo tests to check the sensitivity of the baseline results to violations of key assumptions. We will conduct a few such tests too:

(a) Notice that the several states have both primary and secondary belt laws. Briefly describe how this could be a potential concern to your DID analysis above (hint: reflect back to your finding in part 1.b)?

(b) Estimate the event-study by controlling for state-specific linear trends in the BJS imputation estimator. Do the results seem robust?

(c) Estimate the event-study with time-varying control for vehicle-miles traveled per capita (use the BJS estimator). Do the results seem robust? What could be a potential concern with this control?

(d) Consider parallel trends conditional on the percent of college graduates in 1981 (a time-invariant control). You may use BJS or Callaway and Sant'Anna estimator. Is the event study different than the one you obtained under unconditional parallel trends?